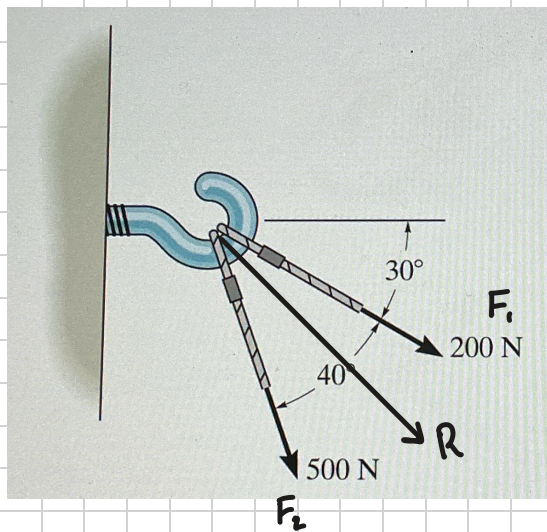


homework



determine the magnitude of R

SOLUTION

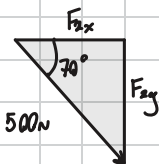
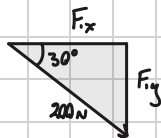
$$F_{1x} = 200 \cos(30^\circ) = 173.21 \text{ N}$$

$$F_{1y} = -200 \sin(30^\circ) = -100.00 \text{ N}$$

(F_{1y} and F_{2y} will be negative)

$$F_{2x} = 500 \cos(70^\circ) = 171.01 \text{ N}$$

$$F_{2y} = -500 \sin(70^\circ) = -469.85 \text{ N}$$



$$R_x = F_1 + F_2 = 344.22 \text{ N}$$

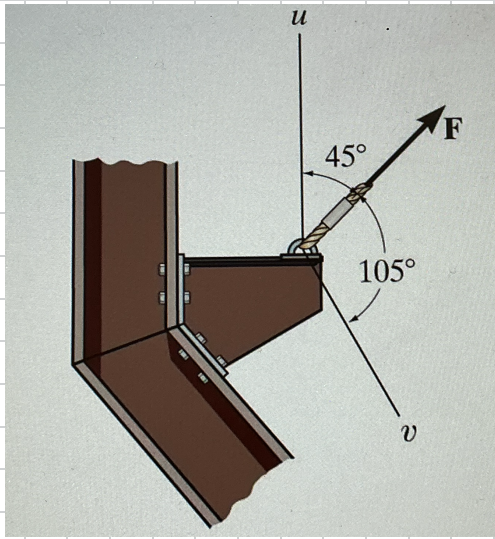
$$R_y = F_1 + F_2 = -569.85 \text{ N}$$

$$R = \sqrt{R_x^2 + R_y^2}$$

$$R = \sqrt{(344.22)^2 + (-569.85)^2}$$

$$R = \sqrt{118,487.4 + 324,729.0} = \sqrt{443,216.4} = 665.75 \text{ N}$$

$\therefore$ the resulting magnitude is 666 N



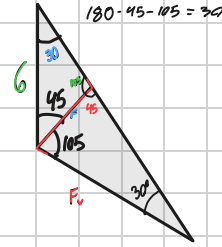
$$F_u = 6 \text{ kN}$$

$$F = 8.5$$

$$F_v = 12$$

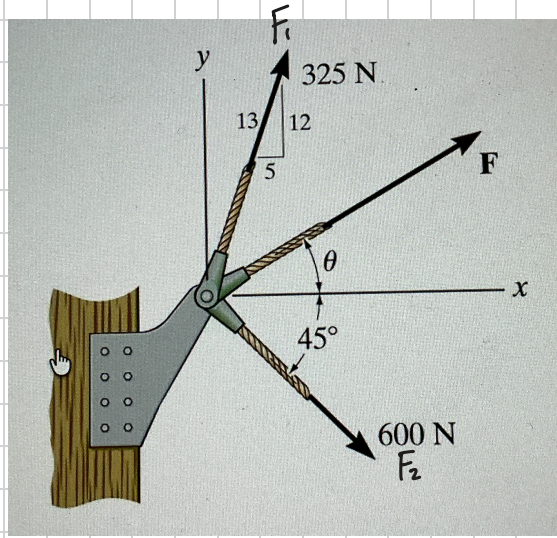
$$\frac{F}{\sin 30} = \frac{6}{\sin 105}$$

$$F = 3.11 \text{ kN}$$



$$\frac{F}{\sin 30} = \frac{F_v}{\sin 45}$$

$$F_v = 4.39 \text{ kN}$$



use not rounded angles

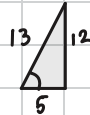
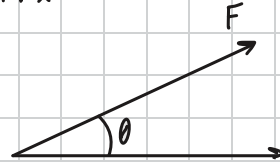
$$F_{1x} = 325 \cos 67 = 126.9876$$

$$F_{1y} = 325 \sin 67 = 299.164$$

$$F_{2x} = 600 \cos 45 = 424.264$$

$$F_{2y} = -600 \sin 45 = -424.264$$

$$\text{net } F_x = 750 \text{ N}$$

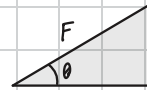


$$\tan^{-1}\left(\frac{12}{5}\right) = \theta$$

$$67^\circ = \theta$$

Find F magnitude and θ

while $F_{\text{net } x} = 750 \text{ N}$



$$\tan \theta = \frac{F \sin \theta}{F \cos \theta} = 0.6294$$

$$\theta = \tan^{-1}(0.6294)$$

$$= 32.187^\circ$$

$$F = \sqrt{(F \cos \theta)^2 + (F \sin \theta)^2}$$

$$= 235 \text{ N}$$

My answers are off by one or two #'s

$$126.9876 + 424.264 + F \cos \theta = 750$$

$$F \cos \theta = 198.7484 \text{ N}$$

$$299.164 - 424.26 + F \sin \theta = 0$$

$$F \sin \theta = 125.096 \text{ N}$$

